

Wireless Low-Cost Luggage Security Alarm



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If someone tries to walk away, say, two metres, with your luggage, this circuit would sound an alarm to indicate that the luggage is being stolen. The alarm mimics a police siren that alerts the user and also scares the thief. As the distance

is very short, the user can catch him and get back the luggage. This device also makes sure that the owner never forgets about the luggage.

Circuit and working

The circuit of the wireless low-cost luggage security alarm has transmitter and receiver sections. Fig. 1 shows the transmitter section while Fig. 2 shows the receiver section.

The transmitter section is built around 5V regulator 7805 (IC1), an encoder HT12E (IC2), a 433MHz transmitter module (TX1) and a few other components. It requires +5V DC power supply, which is obtained from regulator IC1. Switch S1 is used as power on/off switch.

Transmission enable pin 14 of IC2 is grounded as it is an active low input. So, when the circuit is switched on, it starts encoding according to the data given to its input and

address lines. The oscillator is activated by connecting an external 1-mega-ohm resistor (R1) between pin 15 and pin 16 of IC2. DOUT pin of IC2 sends serial data according to the input combination (where A0-A8 equals 11011111 and AD8-AD11 equals 1111) to the RF transmitter module.

Similarly, as shown in Fig. 2, the receiver section is built around 5V regulator 7805 (IC3), a 433MHz receiver module (RX1), tone generator UM3561 (IC5), low-power audio amplifier LM386 (IC6) and a few other components. It also requires +5V DC power supply, which is obtained from regulator IC3. Switch S2 is used as power on/off switch.

The receiver section consists of an RF receiving antenna that receives serial data sent by transmitter TX1 as

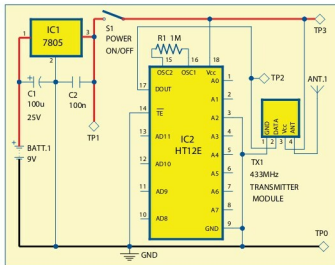


Fig. 1: Circuit of the transmitter section

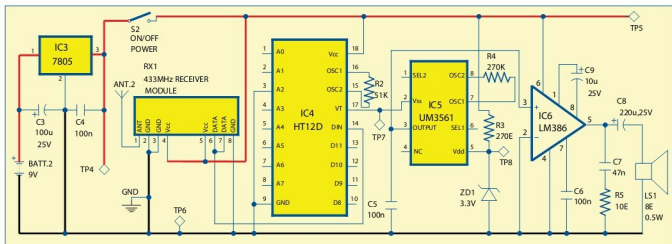


Fig. 2: Circuit of the receiver section

Test Points

Test point	Details
TP0, TP6	0V (GND)
TP1, TP4	5V
TP2	Pulses during transmission
TP3	5V when switch S1 is closed
TP5	5V when switch S2 is closed
TP7	High during transmission and low when there is no transmission
TP8	3.3V